

NAME: K. PAVAN DEEKSHITH

Reg no: 192412172

Sub code: ITA0502

Sub name: Computer vision

Activity: 2

Scenario Based

- ① CCTV image with Salt - and Pepper noise?
 - Technique: Median Filter
 - Reason: Median filtering is a non-linear spatial filter specifically effective at removing impulsive (salt - and pepper) noise without blurring sharp edges.
- ② Satellite image corrupted by Gaussian noise
 - Technique: Gaussian Filter
 - Reason: Gaussian noise is additive and normally distributed.
- ③ medical x-ray image:
 - Technique: (Photographic operations) Bilateral Filter
 - Reason: Bilateral filtering considers both geometric distance and Pixel intensity. Differences effectively smoothing random noise.
- ④ Scanned document with scattered black and white dots
 - Technique: Morphological operation.
 - Reason: Morphological opening removes small white foreground spots.
- ⑤ Low-Light image with heavy grain and no-clear boundaries
 - Technique: Non-local means (NLM) denoising
 - Reason: Averages similar patterns across the entire image to clear heavy grain

⑥ Foggy road image with very low contrast

- Technique: Dark Channel Prior.
- Reason: Removes atmospheric haze and boosts contrast.

⑦ Underwater image that appears dull and washed out

- Technique: CLAHE
- Reason: Enhances local contrast without over-amplifying noise

⑧ Chest x-ray with poor visibility of internal structures

- Technique: Contrast stretching / CLAHE
- Reason: Expands dynamic range to highlight subtle tissue and bone details.

⑨ Night-time traffic image lacking intensity variation

- Technique: Gamma Correction.
- Reason: Brightens dark shadows while keeping headlights from blowing out.

⑩ A grayscale image uses only a narrow intensity range

- Technique: Contrast stretching
- Reason: Spreads narrow intensity values across the full range (0-255) for better clarity